Assignment 3 — Options, Option Risk, and Portfolio Construction

FinTech 545 — Quantitative Risk Management

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Instructions

Be verbose. Explain clearly your reasoning, methods, and results in your written work. Write clear code that is well documented. With 99% certainty, you cannot write too many code comments.

Written answers are worth 8 points. Code is worth 2 points. 10 points total.

1. When finished, respond to the question in Canvas as "done." We will record your grade there.
2. In your code repository, create a folder called "Assignment3."
3. In that folder, include

    a. a document (PDF) with your responses.
    b. All code.
    c. A README file with instructions for us to run your code.

Data for the problems is in the CSV files alongside this document in the class repository.

This assignment covers Weeks 1 through 7, with most of the weight on Weeks 5, 6, and 7. Expect to use the earlier weeks throughout. Moments, distribution fitting, AICc, regression, delta normal VaR, and Expected Shortfall all come back.

Unless a problem says otherwise, use arithmetic returns, and report VaR and ES as positive numbers for a loss. Write your own option pricing code: the generalized Black Scholes Merton formula, its Greeks, and the binomial tree.

How These Problems Are Graded

Every problem below is in three parts, and the middle part is the one you are most likely to skip.

1. **Predict** – look at the data, or a plot of it, and say what you expect and why. Do this before you fit anything.
2. **Fit** – run the models.
3. **Reconcile** – say whether the fit agreed with you. If it did not, say what you had missed.

Most of the written credit sits on parts 1 and 3. Correct code with no reasoning attached does not earn full marks. A wrong prediction, honestly reported and then explained, earns nearly all of them. There are no points deducted for wrong predictions. We are grading whether you understand what the numbers mean, not whether you can call a library function.

1 Implied Volatility from a Dirty Chain (1.5 points)

`problem1.csv` holds the bid and ask for one expiry of European calls and puts on the stock `XYZ`, at 18 strikes from 140 to 225. Use:

- Current price of `XYZ`: $182.40
- Current date: 11/13/2026
- Expiration date: 12/18/2026
- Risk free rate: 4.25%
- Continuously compounded dividend yield: 0.60%

Measure time to maturity in calendar days over 365. Value each quote at its mid price unless a part says otherwise. A market data feed is rarely clean, and this one is not.

Predict.

a. Before computing any implied volatility, check put call parity at every strike using the mid prices. A gap that is smaller than the bid-ask spread cannot be traded, so flag a strike only when the gap exceeds the call's spread plus the put's spread. Report the gap and that band at each strike. What do you conclude about any strike you flagged?
b. If parity holds at a strike, what must be true of the call's and the put's implied volatility there? Answer from the parity relation itself.
c. Implied volatility is found by inverting the price. Using vega, say which quotes you expect to give the least reliable implied volatilities, and why. Look at the quotes in the far wings before you answer.

Fit.

d. Compute the implied volatility of every quote from its mid, and also from its bid and from its ask. Some quotes may not have an implied volatility at one of the three. If so, say why. Plot the call and the put implied volatility against strike on one chart.
e. Compute vega at each strike, and the change in implied volatility that a one cent change in price produces.

Reconcile.

f. Did the strike flagged by parity show up in the plot? Translate its parity gap into vol points using vega, and check that against the gap between its call and put implied volatility.
g. Describe the call wing above 210. Is that shape a statement about the market, or about the quotes? Use your answer to part (e) and the bid and ask implied volatilities to decide, and explain why parity did not catch it.
h. Ignoring the two problems you found, describe the shape of the smile. What does it say about demand for downside protection on `XYZ`?

2 American Options and Discrete Dividends (1.5 points)

A stock trades at \$48.50. Consider a call and a put, both struck at \$50, with 0.5 years to maturity. The risk free rate is 4.5% and the implied volatility is 30%. The stock pays two discrete dividends before expiration:

- \$0.25 at $t = 0.10$ years.
- \$1.50 at $t = 0.40$ years.

Use a binomial tree whose steps fall exactly on the dividend dates, so N is a multiple of 5.

Predict.

a. Apply the condition from Week 7, $D_n > X\left(1 - e^{-r(T-t_n)}\right)$, to each dividend. Just before which dividend, if either, can early exercise of the American call be optimal?
b. Without computing anything, predict for each of the four options (European call, American call, European put, American put) whether the dividends raise or lower its value, and which of the two early exercise premiums, call or put, the dividends make larger.

Fit.

c. Price all four options without dividends, using GBSM for the Europeans and a tree for the Americans. Then price all four with the dividends, using the non recombining tree. For the European options, also report GBSM with the present value of the dividends taken off the stock price, and compare it with the tree.
d. Show how the American values converge as N grows. Include neighbouring values of N, such as 100 and 105.
e. Compute the delta and the gamma of the two American options by central finite differences on the tree, with bump sizes of 1, 0.1, 0.01, and 0.001.
f. Compute the sensitivity of each option's value to each dividend amount.

Reconcile.

g. Report the early exercise premium of each option, with and without dividends. Did your predictions in parts (a) and (b) hold? For the call, find the stock price above which exercise just before the second dividend is optimal.
h. One of your Greeks in part (e) does not settle down as the bump shrinks. Which one, and why? Think about what the tree's value looks like as a function of the stock price when N is fixed.
i. The two dividends move the call by very different amounts per dollar. Explain why, using part (a).

3 VaR of an Option Book (2 points)

`problem3.csv` holds 501 daily prices for the stock `ABC`. `problem3_portfolios.csv` holds four books of positions in `ABC` and European options on it. `Holding` is a number of shares, or a number of options on one share each, and negative means short. Use:

- Value every position at the last price in `problem3.csv`.

- Risk free rate of 4.25%. `ABC` pays no dividends.
- 255 trading days a year. `DaysToExpiry` is in trading days.
- Hold each option's implied volatility constant over the horizon.
- The horizon is 10 trading days. Assume zero expected return.

Predict.

a. Compute each book's delta, gamma, and theta. Before you simulate anything, rank the four books by how badly you expect a delta normal VaR to misstate their 10 day risk, and say whether it overstates or understates. Use the Greeks to justify the ranking.
b. For each book, predict the sign of the P&L if the price does not move for 10 days. Then predict which books will have an ES to VaR ratio well above the 1.25 a normal P&L would give.

Fit.

c. Compute the first four moments of the returns. Fit a normal and a Student's t and choose between them with AICc. Report the fitted parameters.
d. Compute the 10 day delta normal VaR and ES for each book at 5% and 1%.
e. Simulate 10 daily returns from each fitted distribution with zero mean, compound them, and revalue every position in full with 10 fewer days to expiry. Report the mean P&L, and VaR and ES at 5% and 1%, for each book under each distribution.
f. Plot each book's 10 day P&L against the price of `ABC` in 10 days, with the delta only approximation on the same axes.

Reconcile.

g. Two books have a delta normal VaR close to zero. Explain what the delta normal method cannot see, using the second order expansion $\Delta P \approx \delta\,\Delta S + \frac{1}{2}\Gamma(\Delta S)^2 + \Theta\,\Delta t$. Does adding the gamma term reproduce the full revaluation numbers?
h. Compare the mean P&L from part (e) with your prediction in part (b). What is the short gamma books' positive mean paying for?
i. Switching from the normal to the t moves some numbers far more than others. Which numbers, in which books, and why does the 5% VaR move least?
j. The hedged short call has no first order exposure. A risk manager says it is "flat." Write two sentences you would send in reply.

4 A Sharpe Ratio That Sells Insurance (1.5 points)

`problem4.csv` holds 240 monthly total returns for four assets: an equity index `EQ`, a bond index `BOND`, a single stock `STK`, and `PUTW`. Each month, `PUTW` sells a one month put on `EQ` struck 5% below the current level. It holds the strike in cash at the risk free rate, so it never borrows. The risk free rate is 3% a year.

Predict.

a. Compute the first four moments of each asset, and its annualized Sharpe ratio. If one number looks out of place, check how much of it comes from a single month.

b. Before optimizing, predict how much weight a long only maximum Sharpe ratio portfolio will give `PUTW`. Then predict whether that portfolio's historical ES at 1% will be close to the ES of a normal with the same mean and standard deviation.

Fit.

c. Find the maximum Sharpe ratio portfolio three ways: unconstrained, long only, and long only with `PUTW` capped at 20%. Plot the long only efficient frontier and the capital market line, with the four assets marked.
d. For each of the three portfolios, report the Sharpe ratio, and the historical and the normal VaR and ES at 5% and 1%.
e. Run the CAPM regression of `PUTW`'s excess return on `EQ`'s excess return. Report α, its t statistic, β, and R^2. Then fit β separately on the months when `EQ`'s excess return was below -5%.

Reconcile.

f. What has the optimizer bought? Explain why the Sharpe ratio cannot see it, and use the coherence properties from Week 5 to say why ES can.
g. Is `PUTW`'s CAPM alpha evidence of skill? Use the two betas from part (e) to explain what the single regression got wrong.
h. Compute the standard error of each asset's mean monthly return. How precisely do 240 months pin down the inputs to the optimizer? Use that to say which of the tangency weights you would trust. Resampling the months is one way to check.

5 Alpha or Factor Exposure? (1.5 points)

`problem5.csv` holds five years of daily returns, as decimals, for two funds, `FUND1` and `FUND2`. `F-F_Research_Data_Factors_daily.CSV` and `F-F_Momentum_Factor_daily.CSV` hold the Fama French factors and the risk free rate. Fama and French store percentages, so divide by 100. Both funds charge a fee for alpha.

Predict.

a. Compute each fund's mean excess return, volatility, and Sharpe ratio, and fit the CAPM. Which fund looks more skilled?
b. Compute the correlation of each fund's CAPM residual with `SMB`, `HML`, and `Mom`. Predict what will happen to each fund's alpha when those factors are added to the regression.

Fit.

c. Fit the CAPM, the Fama French three factor model, and the Carhart four factor model to each fund. Report each coefficient with its standard error, the alpha's t statistic, R^2, and AICc.
d. Using the four factor betas, compute each fund's expected annual return under the average factor returns of the most recent 10 years in the factor files. Report it with and without the fitted alpha.

e. Find the worst daily momentum return in the factor file. Using the four factor betas, compute the one day loss on $10,000,000 in each fund if it happened again with the other factors flat. Express each loss as a multiple of that fund's daily standard deviation in the sample.

Reconcile.

f. Did the alphas move as you predicted? Use the factor premia during the sample to account for the change in `FUND1`'s alpha.
g. Use Arbitrage Pricing Theory to explain what an alpha is relative to a factor model. Which fund would you pay a fee for alpha, and which model would you use to decide? Label your answer as an opinion where it is one.
h. Compare part (d) with each fund's realized return in the sample. What does that say about building expected returns for an optimizer, like the one in Problem 4, from a factor model?

6 Code (2 points)

Your code is graded on whether we can run it and whether we can follow it. Include a README with the commands to reproduce every number in your write up. Comment the non-obvious steps, particularly anywhere you chose one convention over another.